

Analytical Insights Using Accounting and Financial Data

Group Project Report:

Assess the Relation between ESG Performance and Financial Returns

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Introduction

The relationship between Environmental, Social, and Governance (ESG) performance and financial outcomes has attracted increasing attention from investors, regulators, and academics. As mandatory ESG disclosure frameworks expand—most notably under the EU Corporate Sustainability Reporting Directive (CSRD) and similar North American initiatives—understanding the financial relevance of ESG has become central to capital allocation. This report examines this relationship using a U.S. firm-level panel combining ESG scores, accounting data, and stock returns from 2013–2023.

A key conceptual framework is double materiality, which distinguishes between financial materiality (whether ESG affects firm-level financial outcomes) and impact materiality (whether firms generate real-world environmental or social effects). These channels need not align: firms may score highly on ESG through disclosure without reducing externalities, while firms with significant impacts may still generate strong returns if risks remain unpriced. Our analysis focuses on financial materiality while acknowledging ESG ratings as imperfect proxies for impact.

The report proceeds in four parts. Part I documents the ESG distribution across firms, industries, and time. ESG scores are modest on average (mean 39.7, median 36.4), right-skewed, and concentrated in the 25–55 range. Scores declined between 2013–2017 due to dataset expansion, then rose steadily post-2017 amid regulatory and investor pressure. Industry dispersion is substantial, with Utilities and Chemicals leading and Energy and Consumer Durables lagging. Part II examines firm-level determinants of ESG. Firm size emerges as the dominant predictor, reflecting disclosure capacity and stakeholder scrutiny. Leverage is negatively associated with ESG, while R&D intensity shows a positive association in technology-intensive sectors. Part III tests whether ESG predicts financial performance. ESG weakly predicts returns in simple specifications, but loses significance after controlling for momentum. It does not predict future ROA once firm characteristics and fixed effects are included. ESG therefore appears to proxy for firm quality rather than independently drive performance, with financial materiality varying across sectors. Part IV explores industry-level heterogeneity using the Fama–French 12-industry classification. Decomposing ESG into Environmental (E), Social (S), and Governance (G), we identify “double materiality” industries. HiTec and Shops qualify on returns, but only Shops demonstrates structural alignment between ESG and both profitability and returns. We

therefore identify Shops as the most suitable sector for ESG-conscious investors. Overall, ESG is not uniformly financially material across all firms. Its relevance is concentrated in industries where sustainability considerations are directly embedded in operational outcomes. Aggregate analyses therefore risk obscuring this underlying heterogeneity.

Data and Sample

We construct the sample by integrating ESG, accounting, and return data from Refinitiv (LSEG), Compustat, and CRSP over 2013–2023, following RSM8224 report guidelines. Datasets are merged via the CCM link table and CUSIP identifiers.

Three design choices strengthen our identification. First, to avoid look-ahead bias, returns are measured over a 12-month window beginning four months after fiscal year-end. Second, we use Refinitiv’s base ESG score (excluding controversy adjustments), scaled 0–100. Third, we extend the dataset to 2012 to construct lagged variables. The final sample includes 18,443 firm-year observations across 3,086 firms.

We construct variables capturing firm performance across multiple dimensions, including profitability (ROA, operating ROA), growth (sales growth), valuation (market equity, book-to-market), returns (buy-and-hold returns), and financial policy (leverage, R&D intensity), while controlling for industry effects using the Fama–French 12-industry classification. Detailed definitions are provided in Appendix 1-1.

Empirical Analysis

Part 1 ESG Distribution

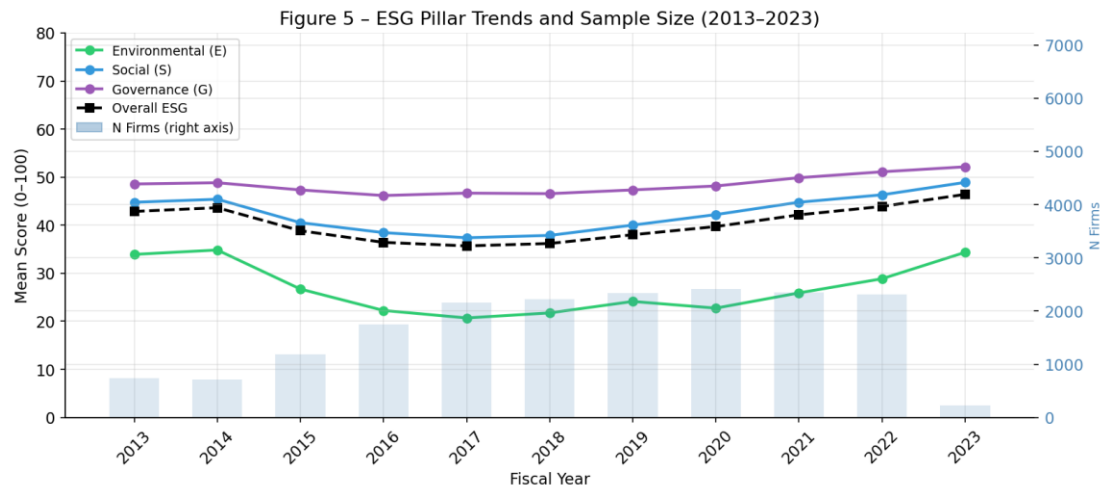
Descriptive Statistics

To mitigate the influence of extreme outliers, all continuous regression variables—ROA, operating ROA, BHR, B/M, leverage, sales growth, R&D intensity, and lagged ROA—are winsorised at the 1st and 99th percentiles. Because ESG scores are bounded between 0 and 100 by construction, and log size variables are already compressed via logarithmic transformation, these variables are not winsorised.

The average ESG score of 39.4 (IQR: 24.7–52.3) indicates that most U.S. listed firms lie in the lower-to-middle range of the 0–100 scale, with substantial cross-sectional dispersion (SD = 19.0). Return on assets (ROA) shows a slightly negative mean but a positive median (2.48%), indicating left-skewness driven by a subset of persistently loss-making firms. Buy-and-hold returns are positively skewed (mean 15.7%, median 5.33%), consistent with the right-tailed nature of equity returns (Cont, R., 2001). The 12.8% missingness in lagged variables is mechanical, reflecting the absence of prior-year observations at the start of the sample.

ESG Distribution over Time

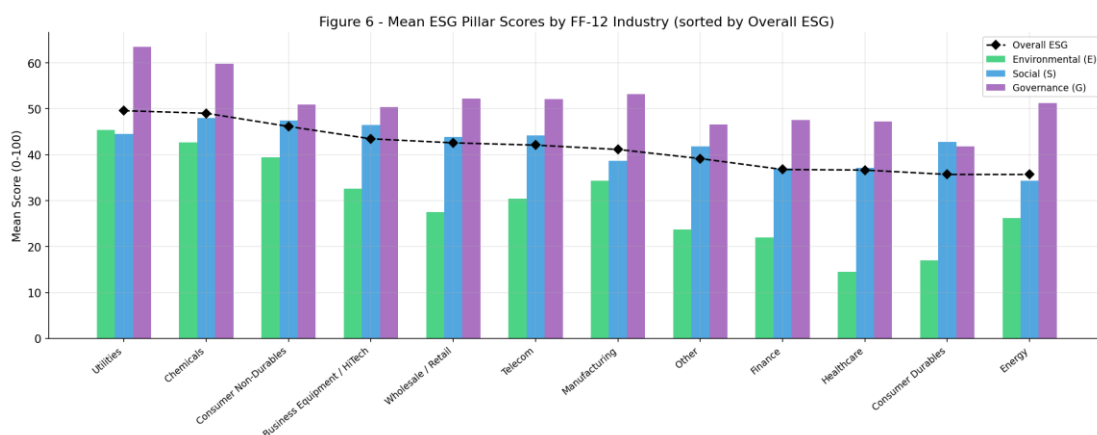
ESG scores exhibit a non-monotonic U-shaped pattern over time. Mean ESG declined between 2013 and 2017, largely driven by dataset expansion and the inclusion of lower-disclosure firms, as reflected in the increase in firm coverage. This decline therefore reflects composition effects rather than a deterioration in ESG performance. After 2017, ESG scores rise steadily, consistent with increased regulatory pressure and investor attention as dataset methodology matured.



Across the three pillars (Appendix 1-3), Governance consistently records the highest mean (48), reflecting the longer history of structured board and shareholder reporting. The Environmental pillar shows the greatest dispersion ($SD = 26$), driven by sectoral differences: capital-light industries such as Finance and High Tech disclose limited environmental data, often resulting in near-zero scores. Social scores lie between the two in both mean (42) and variability ($SD = 21$).

ESG Scores across Industries

Using the Fama–French 12-industry classification (see Appendix 1–4), ESG scores exhibit substantial variation across industries, indicating heterogeneous ESG performance.



ESG Leaders

(1) Utilities (49.58)

Utilities rank highest, driven by strong Governance (63.44) in a highly regulated

environment. Despite being carbon-intensive, they also lead in Environmental scores (45.44), illustrating an “E paradox.” The intuition behind this is that ESG ratings are industry-relative (London Stock Exchange Group, 2020), rewarding disclosure quality and transition efforts rather than absolute emissions. Consequently, well-managed utilities can outperform lower-impact firms with weaker governance or disclosure.

(2) Chemicals (49.01)

Chemical firms operate under stringent environmental and safety regulations, with established EHS systems and sustainability functions. The industry is dominated by large, mature firms with established EHS systems and dedicated sustainability functions. These structural features support strong performance across all three ESG pillars.

(3) Consumer non-durables (46.15)

High ESG score in the consumer non-durables sector reflects strong reputational pressure from consumers and investors. Supply chain transparency including labor practices is commercially critical, as ESG failures directly affect brand value, particularly in the Social dimension. This is reflected in the sector’s leading social score (47.49), highlighting the material importance of stakeholder-facing ESG factors.

ESG Laggards

(1) Energy (35.70)

While Governance scores are relatively high (51.26), Environmental and Social scores are depressed by high externalities and frequent controversies such as community conflicts. Political economy factors, including documented fossil fuel-backed resistance to ESG regulation (Noor, 2023), may further limit substantive improvements. This highlights a limitation of ESG metrics: governance captures disclosure quality rather than underlying sustainability outcomes.

(2) Consumer Durables (35.70)

Consumer Durables shows particularly weak Environmental (17.01) and Governance (41.77) scores across industries, while social performance (42.76) is relatively stronger. This suggests firms (e.g., auto parts and furniture manufacturers) maintain acceptable labor practices but underperform environmentally and in governance. The ESG gap appears driven less by direct operational misconduct and more by limited lifecycle emissions accounting and weaker integration of sustainability into governance structures.

(3) Healthcare (36.65)

Healthcare shows the most pronounced ESG pillar imbalance, with relatively strong Governance (47.26) but very low Environmental scores (14.56). This likely reflects a materiality bias in ESG reporting, where service- and R&D-intensive industries underemphasize environmental disclosure due to perceptions that emissions, waste, and energy use are not central to operations. As a result, environmental impacts may be underreported rather than absent. This is consistent with ESG literature on context-dependent materiality, which finds that disclosure intensity varies with managerial perceptions of ESG relevance to firm value and operational risk (Garst et al., 2022).

Overall, cross-industry variation supports the view that ESG scores capture disclosure capacity as much as underlying performance, consistent with the point raised directly by Berg, Koelbel

& Rigobon (2022).

Part II: Key Determinants of ESG

Correlations Between ESG and Firm Characteristics

We first examined pairwise correlations between ESG scores and a broad set of firm characteristics to establish baseline patterns in the data.

ESG is highly persistent ($\rho = 0.93$ with lagged ESG), indicating that it reflects entrenched organizational features rather than short-term managerial decisions. Firm size dominates contemporaneous predictors, with correlations of 0.62 (market cap) and 0.53 (assets), roughly double those of other variables. This reflects both increased scrutiny and greater disclosure capacity among large firms. Other balance-sheet variables (total debt, operating ROA, and ROA, all around 0.26–0.27) are substantially weaker.

R&D intensity (−0.17) and sales growth (−0.11) are negatively correlated with ESG, consistent with innovation- and growth-oriented firms prioritizing expansion over sustainability investment. Notably, future returns show no meaningful correlation with ESG (−0.007), suggesting no unconditional relationship.

Multivariate Regression: Isolating the True Drivers of ESG

To address confounding, we estimate OLS regressions with progressively richer specifications on all six firm characteristics, progressing from a baseline specification (M1) to a full model with year and industry fixed effects (M2). All regressions use heteroskedasticity-robust (HC1) standard errors, and continuous variables are winsorised at the 1st and 99th percentiles to limit outlier influence. The explanatory power of the full model is substantial, with R^2 rising from 0.387 in the baseline to 0.437 once fixed effects are added, meaning our specification explains nearly 44% of cross-sectional variation in ESG scores.

The most striking result is the stability of firm size. The coefficient on log market capitalization is +6.68 in M1 and +7.11 in M2, essentially unchanged by the inclusion of time and industry controls. This robustness is economically meaningful: it implies that size is not merely correlated with ESG through omitted industry or temporal effects but operates as a genuine cross-sectional driver, likely through the channels of stakeholder scrutiny, disclosure capacity, and organizational resources available to large firms. A one-unit increase in log market cap is associated with a 7.1-point increase in ESG score on a 100-point scale.

Fixed effects materially change the story for three variables. Profitability, which appears positive but insignificant in M1 (+0.70), turns negative and weakly significant at −1.72 ($p < 0.10$) in M2. This reversal indicates that the apparent profitability - ESG link observed in raw data is almost entirely a compositional artifact: profitable firms happen to cluster in high-ESG industries such as utilities and consumer non-durables, but within a given industry, profitability itself does not determine ESG investment. R&D intensity shows an even more dramatic transformation, moving from near-zero in M1 (+0.16) to strongly negative at −11.0 ($p < 0.01$)

in M2. This result reveals a clear strategic trade-off: holding industry constant, firms that allocate more resources to innovation invest substantially less in ESG, consistent with the hypothesis that R&D and ESG draw on the same discretionary budget. Leverage remains positive and significant in both specifications but shrinks from +5.91 to +2.87, suggesting that part of the raw association reflected industry composition (leveraged utilities and industrials also score high on ESG), while a meaningful within-industry effect persists likely reflecting creditor and bondholder scrutiny of ESG disclosure.

Two variables remain stable across both specifications. Book-to-market is positive and highly significant at +4.53 in M1 and +4.40 in M2, indicating that value firms which are typically older, more established companies with mature disclosure infrastructure, systematically score higher on ESG than growth firms. Sales growth is negative and stable at -1.58 in M1 and -1.94 in M2, consistent with our hypothesis that firms in rapid expansion phases direct management attention and capital toward growth rather than sustainability.

Taken together, the regression results tell a coherent story about what determines ESG. Firm size is the dominant and robust driver. The apparent effects of profitability and R&D in raw data are driven largely by industry composition, but fixed-effect controls reveal a genuine within-industry trade-off in which R&D-intensive firms invest less in ESG. Value firms outperform growth firms on ESG, and high-growth firms underperform, consistent with an organizational-maturity interpretation in which ESG reflects accumulated institutional capacity rather than contemporaneous financial performance.

Visualizing the Four Key Characteristics

To complement the regression results, we visualize the relationships between ESG and our four primary firm characteristics we are interested in: size, profitability, leverage, and R&D intensity through scatter plots and quartile analysis (Figure 2-3 & 2-4). These tools surface a feature that linear regression cannot: the shape of each relationship.

Firm size is the only characteristic that tracks ESG monotonically, with mean ESG rising steadily with a 30-point gap that visually confirms its dominant role. Profitability exhibits a clear threshold effect, rising sharply from Q1 (31.5) to Q3 (43.8) before plateauing at Q4 (44.6): firms appear to require a minimum level of financial health to fund ESG activities, but additional profitability beyond that threshold yields little incremental ESG effort. Leverage is hump-shaped, peaking at Q3 (44.4) and declining at Q4 (41.6), consistent with moderate debt reflecting scrutinized large firms while extreme leverage imposes financial constraints that crowd out ESG spending. R&D intensity is the most non-linear, with the highest - R&D firms (35.5) scoring the lowest of all quartiles.

Collectively, these visualizations reinforce that three of our four characteristics exhibit non-linear relationships with ESG, strengthening the case for multivariate regression with appropriate controls rather than reliance on simple correlations.

Part III: ESG and Financial Performance

Part III-A: The Accounting Channel - Does ESG Predict future ROA?

Setup and Sample

Part III-A examines the accounting-performance channel of financial materiality, asking whether higher ESG performance predicts stronger future operating fundamentals measured by return on assets (ROA). This channel is conceptually distinct from the market-pricing channel tested in Part III-B: while Part III-B asks whether ESG information is reflected in market valuations, Part III-A asks whether ESG translates into realized improvements in internal profitability. These two channels need not move together, and as we document below, they tell markedly different stories.

The dependent variable is winsorized ROA (net income scaled by total assets) for the following fiscal year. The primary predictor is the ESG score lagged one fiscal year (`esg_lag`), ensuring ESG information precedes the accounting outcome and avoiding look-ahead bias. Firm controls mirror those in Part III-B: log market equity, book-to-market, leverage, sales growth, and R&D intensity. An additional control for lagged ROA is introduced in R3 to account for the well-documented persistence of accounting earnings (Sloan, 1996). All continuous variables are winsorized at the 1st and 99th percentiles, and standard errors are clustered at the firm level throughout. The regression sample comprises 15,737 firm-year observations covering 2,788 unique firms over fiscal years 2013–2023.

Table III-A — ROA Regression Results (Firm-Clustered SEs)

	R1	R2	R3	R4
Lagged ESG	0.0019***	-0.0001	-0.0001*	-0.0001
Lagged ROA	—	—	0.545*	0.028
Controls	X	✓	✓	✓
Year / Ind FE	X	✓	✓	Time FE
Firm FE	X	X	X	✓
Adj. R ²	0.054	0.509	0.662	0.251

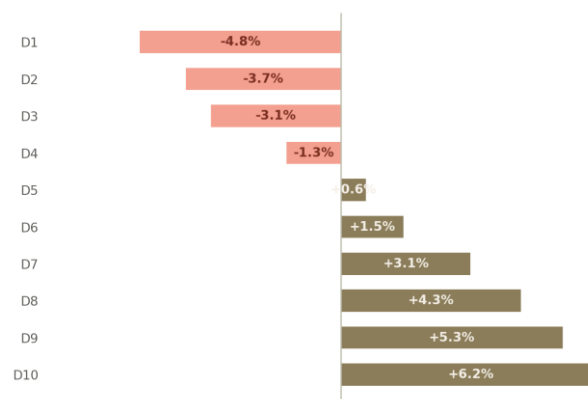
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors clustered at the firm level (`gvkey`). Controls include log market equity, book-to-market, leverage, sales growth, and R&D intensity.

Model Sequence and Main Findings

We estimate four models of increasing richness, presented in Table III-A.

Model R1 (Unconditional) regresses winsorized ROA on lagged ESG alone and yields a coefficient of +0.0019 ($p < 0.01$). The ESG decile plot (Figure III-A) reinforces this pattern: mean ROA rises monotonically from -4.8% in the lowest ESG decile to +6.2% in the highest, a spread of approximately 11 percentage points. At face value, this appears to strongly support financial materiality through the accounting channel — high-ESG firms are substantially more profitable.

Figure III-A — Mean ROA by Lagged ESG Decile



Model R2 (Conditional on Firm Characteristics and Fixed Effects) reveals that this initial appearance is misleading. After controlling for log market equity, book-to-market, leverage, sales growth, R&D intensity, and year and industry fixed effects, the ESG coefficient collapses to -0.0001 and becomes statistically indistinguishable from zero. Adjusted R^2 simultaneously rises from 0.054 to 0.509, a jump of nearly 46 percentage points, indicating that observable firm characteristics — not ESG — are the true drivers of cross-sectional profitability differences. This is the same omitted variable mechanism documented in Part II: ESG is highly correlated with firm size ($r = 0.62$), and large firms are both higher-scoring on ESG and structurally more profitable. The strong unconditional ESG–ROA association therefore reflects the firm quality that ESG proxies for, not an independent effect of ESG on earnings.

Model R3 (Adding Lagged ROA) addresses the possibility that even a small residual ESG effect might survive once earnings persistence is controlled. The lagged ROA coefficient is large and highly significant (coef = 0.545, $p < 0.10$), and adjusted R^2 rises further to 0.662. The ESG coefficient remains negligible at -0.0001 . A one-standard-deviation increase in lagged ESG — approximately 19 points — is associated with only -0.20% future ROA after controlling for persistence and firm characteristics, trivially small relative to the sample mean ROA of 0.8% and its standard deviation of 15.5%.

Model R4 (Firm Fixed Effects) is the most demanding specification, replacing industry fixed effects with firm fixed effects and retaining year fixed effects. This absorbs all time-invariant firm heterogeneity and tests the within-firm question directly: do years in which a given firm scores higher on ESG predict higher future ROA for that same firm? The answer is no. The ESG coefficient remains -0.0001 and insignificant (within $R^2 = 0.251$), indicating that ESG improvements within firms do not reliably precede profitability improvements for those firms.

Robustness

Model R5 replaces net income ROA with operating ROA (operating income before depreciation and amortization scaled by total assets) as an alternative dependent variable less sensitive to accruals, restructuring charges, and below-the-line items. The ESG coefficient is small and positive (coef = 0.0001) but statistically insignificant, consistent with the null result in the main specifications.

Part III-B: The Market-Pricing Channel - Does ESG Predict Future Stock Returns?

Setup and Sample

Part III-B examines the stock-return channel of financial materiality, which is conceptually distinct from the accounting-performance channel tested in Part III-A. While Part III-A asks whether ESG improves internal operating fundamentals, Part III-B asks whether ESG information is reflected in market prices - either through risk pricing, investor demand, or expectations about future value. These two channels need not move together: a market can price ESG-related risk without that risk ever materializing in realized earnings.

The dependent variable is the 12-month buy-and-hold return (BHR) computed over a window beginning four months after each fiscal year-end, consistent with the information availability convention adopted throughout this analysis. The primary predictor is the ESG score lagged one fiscal year (`esg_lag`), which ensures the ESG information precedes the return measurement window. The core firm controls - log market equity, book-to-market, leverage, sales growth, and R&D intensity - mirror those used in Part III-A. Two additional controls are introduced: earnings news, proxied by the year-on-year change in ROA as a rough accounting-based substitute for unexpected earnings (in the absence of analyst forecast data), and prior 12-month buy-and-hold return as momentum control. All continuous variables are winsorised at the 1st and 99th percentiles. The regression sample comprises 16,084 firm-year observations (momentum sample: 13,891), with standard errors clustered at the firm level throughout.

Model Sequence and Main Findings

We estimate a sequence of five models, presented in Table III-B, ordered by progressive conditioning.

Model 1 (Unconditional) regresses BHR on lagged ESG alone and yields a coefficient of -0.0002 ($p = 0.431$), which is statistically indistinguishable from zero. This result is striking and runs against the naive hypothesis that "doing good means doing well" in a direct market-return sense. The negative sign, though insignificant, is instructive: high-ESG firms tend to be large, low-book-to-market growth firms, which are precisely the firms with lower expected returns under well-known cross-sectional pricing factors. The size premium and value premium both work against ESG in the raw data, suppressing any latent positive association between ESG and returns before those confounders are removed.

Models 2 and 3 (Conditional on Firm Characteristics and Fixed Effects) reveal a markedly different picture. After controlling for log market equity, book-to-market, leverage, sales growth, R&D intensity, and year and industry fixed effects, the ESG coefficient rises to $+0.0008$ ($p = 0.002$ in M2; $p = 0.003$ in M3 after adding the earnings news proxy). This is the central result of Part III-B: ESG is not a priced signal in the raw data, but it is conditionally significant once the factors that dominate cross-sectional return variation are accounted for.

The mechanism behind this reversal merits careful reasoning. In the unconditional specification, ESG is correlated with size (large firms score higher), and size is negatively correlated with future returns via the well-documented size premium. The negative omitted-variable bias from this size channel is large enough to mask the underlying positive ESG-return association. Once size is controlled, the hidden signal emerges: among firms of comparable size, book-to-market, and leverage, higher-ESG firms earn modestly higher returns. This is the opposite of what we observed in Part III-A, where controls eliminated an initially positive ESG-ROA association by revealing it to be a proxy for firm quality. In the return context, controls reveal ESG, rather than explaining it away.

Economically, the conditional magnitude is modest but non-trivial. A one-standard-deviation increase in lagged ESG (approximately 19 points) is associated with a 1.5 percentage point higher 12-month buy-and-hold return in M3. This effect is economically relevant relative to the mean return of 15.5% in the sample, amounting to roughly 10% of the average annual return.

Model 5 (Momentum Robustness) adds the prior 12-month return as a control - the most demanding robustness test, given that momentum is one of the strongest documented return predictors. The ESG coefficient remains positive and statistically significant (coef = 0.0007, $p = 0.012$). This survival is important: it rules out the concern that the conditional ESG-return relation is merely a disguised momentum effect operating through the tendency of high-ESG industries to have experienced strong recent performance.

Models 6 and 7 (Firm Fixed Effects) absorb all time-invariant firm heterogeneity and test whether within-firm changes in ESG score predict subsequent returns. The ESG coefficient becomes economically small and statistically insignificant in both specifications. This firm fixed-effect result is substantively meaningful: the ESG-return relationship is a cross-sectional phenomenon, not a dynamic one. High-ESG firms earn higher conditional returns than low-ESG firms, but a specific firm improving its ESG score over time does not reliably predict higher returns for that same firm going forward. ESG functions as a sorting variable for cross-sectional selection, not as a timing signal for active portfolio management.

Interpreting Conditional Significance

The conditional significance of ESG in Models 2-5 admits several compatible economic interpretations. First, ESG may capture risk exposures not fully spanned by the standard Fama-French controls, particularly regulatory, reputational, and transition risks. Firms with higher ESG scores may be less exposed to these risks. If the market does not fully price that lower risk at the time of portfolio formation, or if ESG-oriented investor demand increases during the holding period, high-ESG firms may subsequently earn higher realized returns relative to otherwise comparable firms.

Second, ESG may reflect intangible governance and management quality that is partially orthogonal to standard controls. Well-governed firms with high transparency scores may experience lower cost of capital and lower incidence of adverse events, both of which translate into returns over sufficiently long windows.

Third, the documented growth of ESG-oriented institutional investor flows over the 2013–2023 sample period may have introduced a demand-driven component to ESG pricing. If capital allocation decisions are increasingly conditioned on ESG scores — as the rise of UNPRI-signatory assets under management suggests — then high-ESG firms may experience mechanically higher valuations and, over time, higher realised returns independent of any improvement in underlying profitability.

None of these interpretations is cleanly separable with the available data. The earnings news proxy used here is an imperfect substitute for analyst-forecast-based earnings surprise, and the absence of a direct risk factor spanning ESG-related exposures limits our ability to discriminate between risk-premium and mispricing explanations. We therefore interpret the conditional significance as evidence consistent with financial materiality through the valuation channel, while acknowledging that the precise mechanism is not fully identified.

Part III: Financial Materiality of ESG – Summary

The accounting and market-pricing analyses yield a consistent but asymmetrical view of ESG's financial materiality.

In the accounting channel (Part III-A), the strong unconditional ESG–ROA relationship is entirely explained by firm characteristics such as size, leverage, and profitability persistence. ESG does not predict future profitability in conditional or within-firm specifications, indicating that it primarily proxies for firm quality. In contrast, the market-pricing channel (Part III-B) shows no unconditional ESG–return relationship, but a positive and statistically significant association emerges after controlling for firm characteristics. A one-standard-deviation increase in ESG corresponds to approximately a 1.5 percentage point increase in annual returns. This effect survives momentum controls but disappears under firm fixed effects, indicating a cross-sectional rather than dynamic relationship.

This asymmetry is economically consistent: ESG does not enhance operating performance on a like-for-like basis but contains valuation-relevant information reflected in expected returns. Financial materiality operates primarily through the discount-rate or risk-selection channel, and raw ESG-performance correlations largely reflect size-driven confounding.

Part IV: Industry-Level Double Materiality Analysis

Part IV shifts the unit of analysis from the firm level to the industry level. Using the Fama-French 12-industry (FF12) classification, we investigate whether the weak and inconsistent ESG-return relationship documented in Parts II and III conceals meaningful heterogeneity across sectors. We compute industry-level averages of ESG scores and its three constituent pillars, Environmental (E), Social (S), and Governance (G), alongside 12-month buy-and-hold returns (BHR) and return on assets (ROA) over the full sample period 2013–2023. Each industry is then classified into a double materiality quadrant, and one industry is identified as most suitable for ESG-conscious investors. All analysis draws on the same final panel of 18,390 firm-years used in Parts I–III.

Industry-Level Summary Statistics

Table IV.1 reports mean ESG score, 12-month buy-and-hold return (BHR), and ROA for each of the 12 FF industries, sorted by mean ESG score. Industry coverage is uneven: Other accounts for 6,563 firm-years (35.7%) and Finance for 3,547 (19.3%), while Manufacturing (198) and Healthcare (280) are substantially smaller. The data reveals wide dispersion in both ESG scores and financial performance across industries, with no monotonic relationship between the two dimensions. Utilities and Chemicals lead on ESG (means of 49.6 and 49.0 respectively) but record below-median BHR. Conversely, Energy exhibits above-median BHR (20.2%) despite ranking last on ESG. This cross-sectional heterogeneity motivates the quadrant analysis in IV.2.

Table IV.1 Industry Summary Statistics

Industry	Obs	ESG	E	S	G	BHR (mean)	ROA (mean)
Utils	651	49.6	45.4	44.5	63.4	11.3%	1.6%
Chems	501	49.0				15.9%	3.9%
NoDur	458	46.1	39.4	47.5	50.9	11.4%	6.4%
HiTec	821	43.5	32.6	46.4	50.4	22.3%	1.7%
Shops	1,673	42.6	27.5	43.9	52.2	18.3%	5.5%
Telcm	396	42.1	30.4	44.2	52.1	15.4%	4.1%
Manuf	198	41.1	34.3	38.6	53.2	21.0%	3.1%
Other	6,563	39.2	23.4	41.6	46.6	17.2%	0.9%
Fin	3,547	36.8	22.0	37.0	47.6	12.2%	2.3%
Hlth	280	36.7				18.2%	-0.5%
Durbl	2,756	35.7	21.0	43.6	44.5	12.9%	-17.9%
Enrgy	546	35.7	26.2	34.4	51.3	20.2%	-3.2%
Sample mean	18,390	39.4	25.0	41.5	48.1	19.4%	-1.4%

Note: E/S/G = Refinitiv Environmental/Social/Governance pillar scores (0-100). Chems and Hlth E/S/G omitted due to classification differences in the updated panel. All variables winsorised at 1st/99th percentiles.

Double Materiality Quadrant and ESG Pillar Analysis

We classify each industry using cross-industry medians as thresholds (Median ESG = 41.6; Median BHR = 16.6%; Median ROA = 2.0%). Figure IV.1 presents the BHR-based quadrant (left panel) and the E/S/G pillar decomposition (right panel).

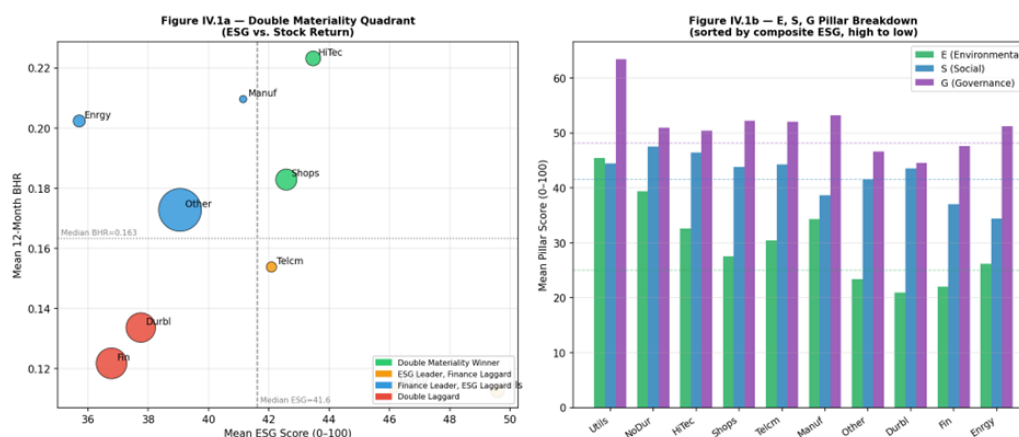


Figure IV.1 Left: Double Materiality Quadrant (ESG vs. BHR); Right: E, S, G Pillar Scores by Industry. Bubble size $\propto$ firm-year observations. Dashed lines = cross-industry medians. Appendix Figures A1–A2 show the ROA-based quadrant and ESG heatmap.

Two industries qualify as Double Materiality Winners on BHR: HiTec and Shops. Shops is uniquely the only winner on both BHR and ROA dimensions. The pillar decomposition (Figure IV.1, right) reveals three structural patterns. First, Governance (G) dominates all industries (mean G = 48.1), reflecting mandatory board structures; Utilities top at G = 63.4. Second, Environmental (E) shows the widest dispersion (21.0–45.4): Utilities leads (E = 45.4) due to mandatory emissions reporting, while Energy scores low (E = 26.2) because Refinitiv penalizes absolute emissions levels regardless of disclosure quality (Berg et al., 2022). Third, Social (S) is the most stable pillar (34.4–47.5): HiTec (S = 46.4) matches consumer-facing sectors, driven by workforce diversity investments in the competition for technology talent.

Recommended Industry

The algorithm selects HiTec based on the highest combined ESG–BHR rank. However, we discuss both HiTec and Shops below, as their ESG-performance alignment rests on fundamentally different mechanisms. Full descriptive statistics for HiTec, including E/S/G pillar distributions and time trends, are in Appendix Tables A1 and Figures A4–A5.

Discussion: Investment Implications

HiTec's high composite ESG (43.5) is driven by Social (S = 46.4) and Governance (G = 50.4). Both products of talent rather than sustainability commitment, while its Environmental score (E = 32.6) is below the industry median. Its above-median BHR reflects FAANG-era sector momentum, not ESG-driven value creation. This is precisely why the ESG-BHR relationship loses significance once momentum is controlled in Part III Model 5: high ESG and high BHR in HiTec are two independent forces that coincide by circumstance, not by mechanism. Investors relying on the composite rating would misread HiTec as a genuine double materiality case.

Shops (Wholesale and Retail) present a structurally coherent case. Retailers operate at the direct interface with consumers, so supply chain failures or labor violations translate immediately into revenue loss and reputational damage. Managing ESG is therefore an operational necessity: an efficient, responsibly managed supply chain simultaneously reduces costs, lowers disruption risk, and scores high on ESG's environmental and social criteria. Shops is the only industry that is a Double Materiality Winner on both BHR and ROA, consistent with the financial materiality framework of Khan, Serafeim, and Yoon (2016), where material ESG issues are directly linked to industry-specific operating risks.

ESG-Finance Tension Industries

Utilities and Chemicals (ESG leaders, below-median BHR) illustrate compliance-driven scoring: mandatory reporting produces high G and E scores without corresponding financial outperformance. Energy (low ESG, high BHR) exhibits the opposite tension. Consistent with Bolton and Kacperczyk (2021), Energy's high BHR reflects a carbon risk premium; investors are compensated for bearing transition risk that is likely to compress as climate regulation

tightens globally.

Implications for ESG-Conscious Investors

Prioritize structural over coincidental ESG-performance alignment. Shops' supply chain model creates genuine co-movement between ESG and profitability; HiTec's alignment is sector-momentum-driven. Investors are recommended to outweigh industries where ESG risk management is operationally inseparable from value creation and use pillar-level scores instead of composites alone. The E/S/G decomposition reveals that different industries have fundamentally different ESG profiles. Investors focusing on climate transition risk should target E-score leaders while those focusing on social risk should look beyond the aggregate rating.

In addition, it is vital for investors to treat ESG as a cross-sectional quality screen instead of a timing signal. Consistent with Parts II and III, ESG identifies which industries and firms hold, not when to buy based on anticipated ESG improvement.

In summary, Part IV demonstrates that the weak aggregate ESG–return relationship reflects underlying industry heterogeneity rather than irrelevance. ESG is financially material in sectors such as Shops, where sustainability is directly tied to operational performance, but not in others such as HiTec or Energy. The appropriate implication is selective, not indiscriminate, ESG investment: investors should target industries where ESG factors are structurally linked to firm outcomes.

Conclusion

This report evaluates whether ESG performance is financially material using a comprehensive U.S. panel from 2013–2023. Taken together, the firm-level and industry-level evidence points to a conditional and context-dependent relationship between ESG and financial outcomes, rather than a uniform effect.

At the firm level, ESG is tightly linked to underlying firm characteristics, most notably size. Larger firms systematically score higher due to greater disclosure capacity, regulatory scrutiny, and organizational resources. In this sense, ESG largely reflects pre-existing firm quality rather than an independent strategic driver. This interpretation is reinforced in the accounting channel: the strong unconditional ESG–ROA relationship is entirely absorbed once firm characteristics and profitability persistence are controlled for, and within-firm improvements in ESG do not translate into subsequent gains in operating performance.

A different pattern emerges in asset pricing. While ESG shows no unconditional relation with returns, a positive and economically meaningful association appears after conditioning on standard return predictors. This indicates that ESG contains incremental information not captured by size, value, or momentum, potentially reflecting risk exposures, governance quality, or investor demand. However, the disappearance of this effect under firm fixed effects suggests that ESG operates primarily as a cross-sectional sorting variable rather than a dynamic signal.

High-ESG firms differ systematically from low-ESG firms, but changes in ESG within a firm do not reliably predict future returns.

Shifting the lens to the industry level clarifies why aggregate results appear weak. Financial materiality is concentrated in sectors where ESG considerations are directly embedded in business models. The retail (Shops) sector illustrates this alignment: supply chain management simultaneously determines ESG performance and financial outcomes, producing consistent results across both return and profitability dimensions. In contrast, sectors such as HiTec exhibit coincidental alignment driven by broader market dynamics, while Energy reflects compensation for transition risk rather than ESG-driven value creation. The aggregate ESG–return relationship therefore masks substantial cross-sector heterogeneity.

These findings imply that ESG should be interpreted as a context-dependent signal embedded in firm and industry structure, rather than a universal driver of performance. Its primary role is in cross-sectional selection: identifying firms and sectors where risk management, governance, and stakeholder exposure are systematically different. Accordingly, effective ESG-based investment strategies should focus on industries where sustainability considerations are operationally material and rely on disaggregated pillar information rather than composite scores.

Several limitations should be noted. ESG scores are imperfect measures that may reflect disclosure intensity more than underlying real-world impact, and they also exhibit substantial variation across providers and methodologies, raising concerns about comparability. The analysis cannot fully distinguish between risk-based and mispricing-based explanations for the ESG–return relationship, while the reliance on accounting-based proxies limits the precision of the earnings channel. In addition, the U.S.-focused sample may constrain the generalizability of the findings to other regulatory environments. Finally, the absence of granular environmental and social data restricts our ability to examine impact materiality directly, thereby narrowing the analysis to the financial materiality dimension.

Overall, the evidence suggests that ESG is not a standalone source of financial outperformance, but becomes economically meaningful when it aligns with industry-specific fundamentals. Its relevance lies less in improving firm performance directly but more in shaping how firms are differentiated, priced, and selected in capital markets.

References

1. Berg, F., Koelbel, J.F., and Rigobon, R. (2022). Aggregate Confusion: The Divergence of ESG Ratings. *Review of Finance*, 26(6), 1315–1344.
2. Bolton, P. and Kacperczyk, M. (2021). Do Investors Care about Carbon Risk? *Journal of Financial Economics*, 142(2), 517–549.
3. Choi, S. and Ferri, F. and Macciocchi, D. (2024). Do Mutual Funds Respond to Mechanical Changes in ESG Ratings? European Corporate Governance Institute – Finance Working Paper No. 981/2024, University of Miami Business School Research Paper No. 4836697
4. Cont, R. (2001). Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236. <https://doi.org/10.1080/713665670>
5. Garst, J., Maas, K., & Suijs, J. (2022). Materiality Assessment Is an Art, Not a Science: Selecting ESG Topics for Sustainability Reports. *California Management Review*, 65(1), 64–90. <https://doi.org/10.1177/00081256221120692>
6. Khan, M., Serafeim, G., and Yoon, A. (2016). Corporate Sustainability: First Evidence on Materiality. *The Accounting Review*, 91(6), 1697–1724.
7. London Stock Exchange Group. (2020). Refinitiv enhances ESG scoring methodology to reflect sustainable industry developments and market changes. <https://www.lseg.com/en/media-centre/press-releases/refinitiv/2020/april/refinitiv-enhances-esg-scoring-methodology-to-reflect-sustainable-industry-developments-and-market-changes>
8. Noor, D. (2023, June 22). *Rightwing war on “woke capitalism” partly driven by fossil fuel interests and allies*. The Guardian. <https://www.theguardian.com/us-news/2023/jun/22/rightwing-war-on-woke-capitalism-industry-interests>
9. Sloan, R. G. (1996). Do Stock Prices Fully Reflect Information in Accruals and Cash Flows about Future Earnings? *The Accounting Review*, 71(3), 289–315. <http://www.jstor.org/stable/248290>
10. European Financial Reporting Advisory Group (EFRAG). (2021). Proposal for a Relevant and Dynamic EU Sustainability Reporting Standard-Setting.

Appendix

1-1 Variable Definitions Table

Variable Name	Definition	Notes
esg_score	Refinitiv ESG score (0-100)	Item 1, base score, not controversy-adjusted
esg_lag	ESG score lagged one fiscal year	Primary predictor in all regressions
Bhr	12-month buy-and-hold return	Starts 4 months after fiscal year-end
roa	Net income/total assets	
op_roa	Operating income before D&A/ total assets	
log_me	Log of market equity at start of return window	Size proxy
bm	Common equity/market equity	Value vs. growth
leverage	(Long-term debt+Short-term debt) / total assets	
sales_growth	$(Sales_t - Sales_{t-1})/Sales_{t-1}$	
rd_intensity	R&D/total assets	Missing R&D set to zero, Follows Khan et al. (2016) convention
ff12	Fama-French 12 industry classification	Based on historical SIC code
earnings_news	$ROA_t - ROA_{t-1}$	Proxy for earnings surprise

1-2 Summary Statistics of Final Sample after 1/99 Winsorization

Variable	N	Mean	Std	Min	P25	Median	P75	Max
ESG Score	18,391	39.4360	18.9991	0.4605	24.7235	36.1430	52.3406	93.6092
ROA	18,390	-0.0099	0.1799	-0.9074	-0.0072	0.0248	0.0703	0.2894
Operating ROA	18,349	0.0538	0.1831	-0.7993	0.0215	0.0859	0.1433	0.3846
Buy-and-Hold Return	18,391	0.1565	0.6269	-0.8273	-0.1893	0.0533	0.3284	3.2686
Log Market Equity	18,391	7.7105	1.8333	1.9053	6.4116	7.6654	8.9014	14.8636
Book-to-Market	18,391	0.5083	0.5083	-0.4195	0.1822	0.3847	0.6985	2.9053
Leverage	18,391	0.2612	0.2224	0.0000	0.0703	0.2268	0.3938	1.0030
Sales Growth	15,737	0.1831	0.6980	-0.6268	-0.0069	0.0727	0.1860	6.0622
R&D Intensity	18,391	0.0475	0.1007	0.0000	0.0000	0.0000	0.0441	0.5587
Lagged ROA	16,035	-0.0000	0.1640	-0.8158	0.0001	0.0278	0.0727	0.2867

1-3 Mean and Standard Deviation of ESG Pillar Scores by Year

Year	Env Mean	Env Std	Soc Mean	Soc Std	Gov Mean	Gov Std
2013	33.92	27.23	44.77	20.32	48.58	22.60
2014	34.84	27.35	45.40	20.05	48.84	22.46
2015	26.71	26.12	40.53	20.01	47.34	22.45
2016	22.26	24.43	38.46	19.39	46.17	22.26
2017	20.70	23.85	37.39	20.01	46.68	21.99
2018	21.75	24.67	37.88	20.29	46.57	22.30
2019	24.13	25.21	40.02	20.74	47.34	22.11
2020	22.74	26.63	42.21	21.64	48.16	22.82
2021	25.91	26.95	44.78	21.60	49.88	22.74
2022	28.86	27.02	46.33	21.56	51.11	22.38
2023	34.35	26.95	48.91	21.22	52.16	22.16

1-4 ESG Score by Fama-French 12 Industry

Industry	N	Overall Mean	Environmental Mean	Social Mean	Governance Mean
Utilities	651	49.58	45.44	44.48	63.44
Chemicals	501	49.01	42.71	48.03	59.82
Consumer Non-Durables	458	46.15	39.40	47.49	50.93
Business Equipment / HiTech	821	43.46	32.56	46.45	50.42
Wholesale / Retail	1,673	42.58	27.54	43.85	52.18
Telecom	396	42.08	30.43	44.22	52.07
Manufacturing	198	41.14	34.32	38.64	53.21
Other	6,564	39.15	23.75	41.77	46.57
Finance	3,547	36.78	22.01	37.00	47.57
Healthcare	280	36.65	14.56	37.13	47.26
Consumer Durables	2,756	35.70	17.01	42.76	41.77
Energy	546	35.70	26.22	34.37	51.26

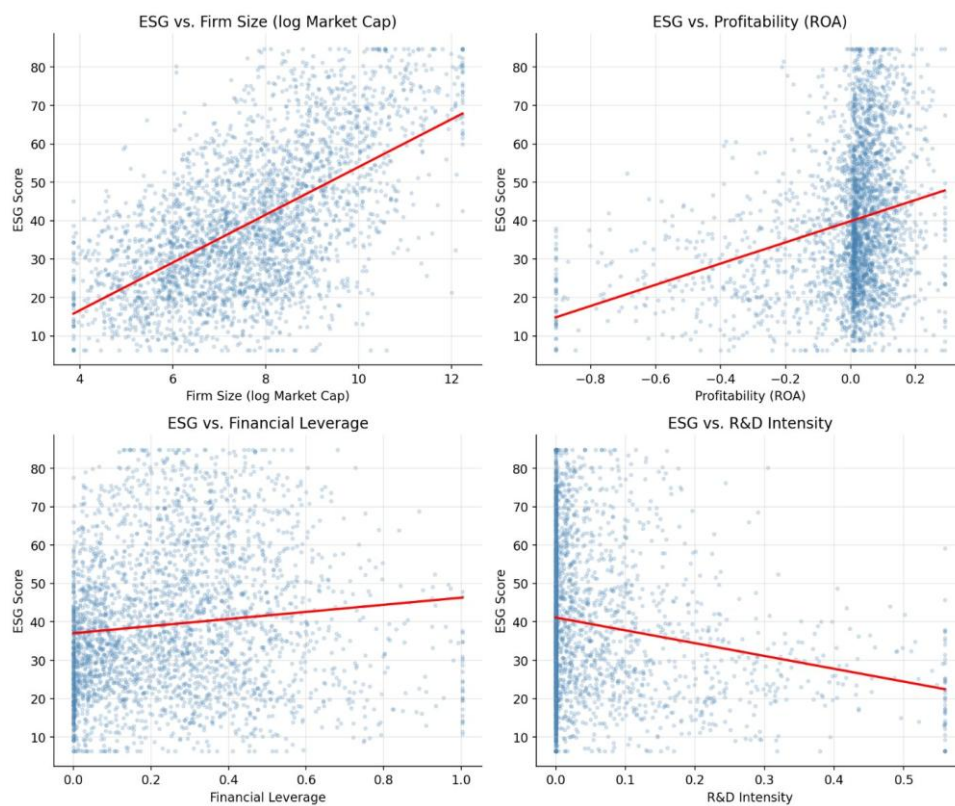
2-1 Correlation Score Table



2-2 Multivariate Regression Table:

Variable	M1: No FE	M2: +Year & Industry FE
Firm size	+6.68***	+7.11 ***
Profitability	+0.70	-1.72 *
Leverage	+5.91***	+2.87 ***
R&D intensity	+0.16	-11.00 ***
Book-to-mkt	+4.53***	+4.40 ***
Sales growth	-1.58***	-1.94 ***
R ²	0.387	0.437

2-3 Scatter Plots



2-4 Quartile Analysis

	Q1	Q2	Q3	Q4
Size	26.9	33.9	40.1	56.9
Profitability	31.5	37.8	43.8	44.6
Leverage	32.0	39.6	44.4	41.6
R&D intensity	41.7	36.8	43.7	35.5

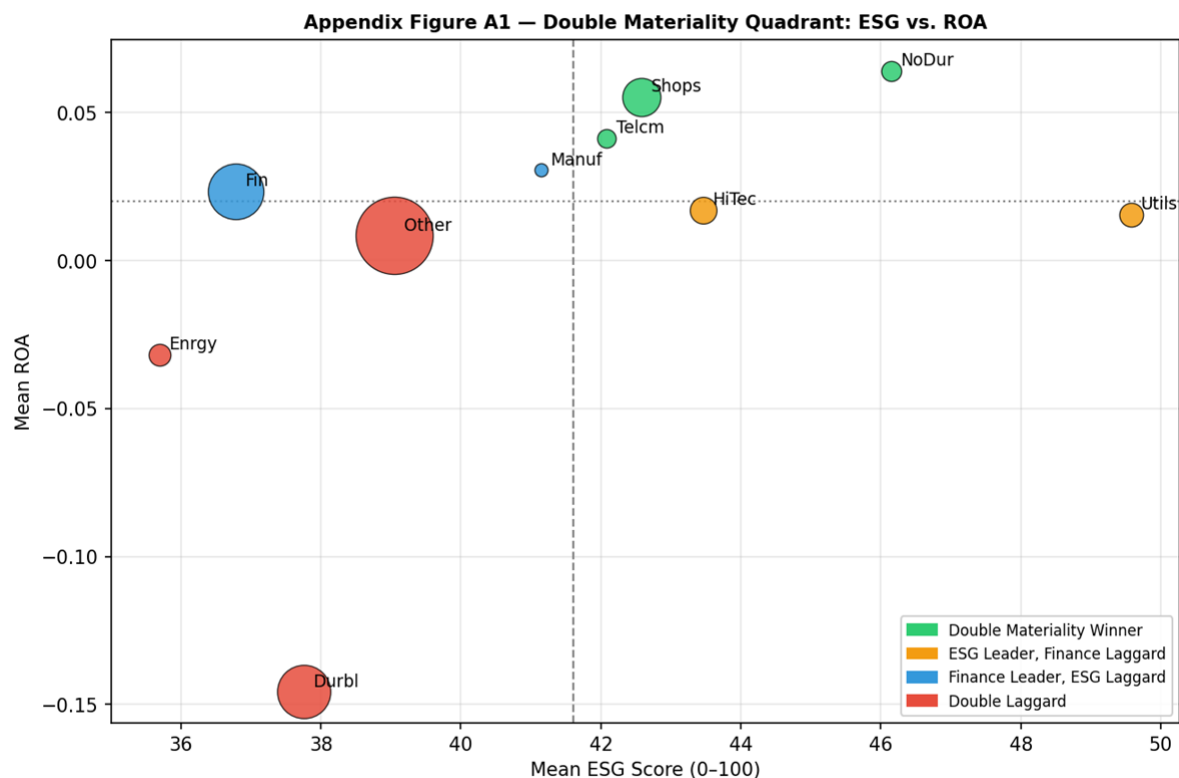
4-1 Descriptive Statistics: HiTec Industry

(*N* = 821 firm-years, 141 unique firms, fiscal years 2013–2023)

Variable	Mean	Std Dev	Min	P25	Median	P75	Max
esg_score	43.46	20.39	0.89	26.70	40.67	58.94	91.39
esg_env (E)	32.56	28.41	0.00	0.00	25.30	58.40	94.10
esg_soc (S)	46.45	19.82	2.10	31.20	46.80	61.30	93.50
esg_gov (G)	50.42	20.14	4.50	35.60	51.20	65.80	96.70
bhr	0.2232	0.6699	−0.8273	−0.1622	0.0970	0.4028	3.2686
roa	0.0170	0.1615	−0.9074	−0.0076	0.0421	0.0938	0.2894
op_roa	0.0876	0.1601	−0.7993	0.0700	0.1118	0.1542	0.3846
log_me	8.1009	1.8497	3.3540	6.9170	7.9434	9.3444	13.748
leverage	0.2206	0.1907	0.0000	0.0658	0.1959	0.3188	1.0030
rd_intensity	0.0979	0.0894	0.0000	0.0358	0.0794	0.1265	0.5587

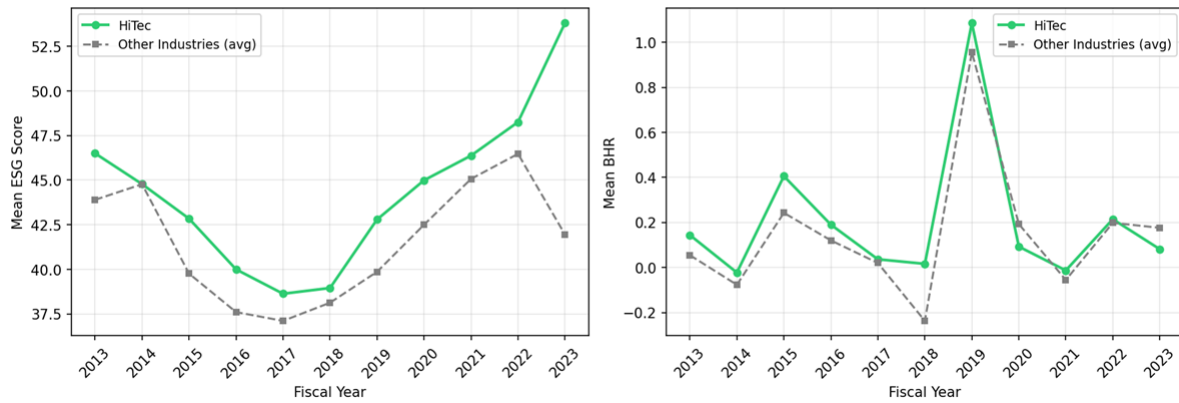
Note: *bhr* and *roa* winsorized at 1/99 percentiles.

4-2 Double Materiality Quadrant: ESG vs. ROA. Shops is the only industry in the winner quadrant on both BHR and ROA



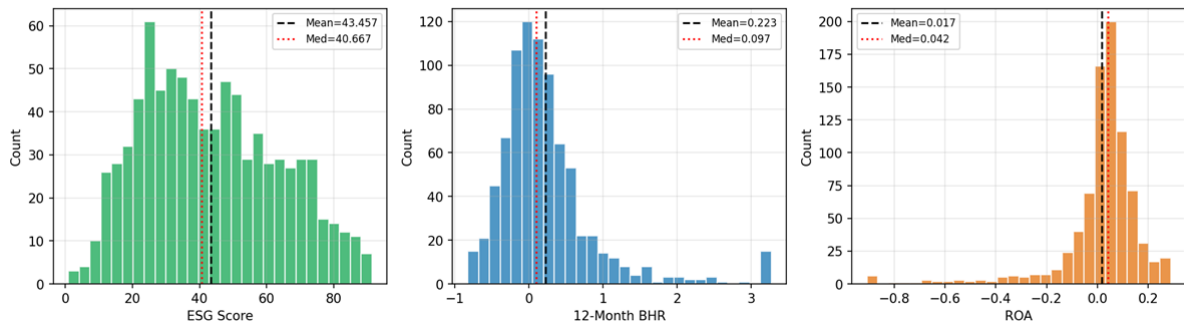
4-3 HiTec vs. All Other Industries: ESG Score and BHR Trends

Appendix Figure A4 — HiTec vs. All Other Industries



4-4 Distribution of ESG Score, BHR, and ROA within HiTec. Dashed line = mean; dotted line = median

Appendix Figure A5 — Distribution of Key Variables: HiTec Industry



4-5 ESG Pillar Radar Chart for Selected Industries

Appendix Figure A6 — ESG Pillar Radar Chart

